Lab Experiment 2

Learning in Engineering and Technology: Processes and Topics

Learning Outcomes

By the end of this lab experiment, students will be able to:

1. Identify and differentiate cognitive, psychomotor, and affective domains of learning through personal examples.
2. Recognize the importance of affective learning, including attitudes, values, and interpersonal skills.
3. Reflect on their own learning preferences and styles to improve learning strategies.
4. Identify common academic challenges and develop plans to overcome them.
5. Reflect on and evaluate their current study habits and identify steps for improvement.
6. Convert measurements between inches/centimeters, feet, and meters.
7. Find and record the current temperature in Fahrenheit, then convert it to Celsius and Kelvin.
8. Use real-world data and apply appropriate formulas for unit and temperature conversions.

Lab exercise

Objective:

To deepen your understanding of engineering and technology through reflection, research, and planning activities.

Instructions:

Complete the following questions from chapters **1** and **2** in your lab report paper.

Chapter 2 - Potentials in Engineering:

- What field of engineering are you interested in and explain why do you want to be an engineer?
- Have you ever had a job you didn't like? Describe the job. What didn't you like about it? If that job played any role in your subsequent decision to major in engineering, explain what that role was.
- What potential do you have to assist in you becoming an engineer?

Chapter 3 - Learning Processes:

Think of the three categories of learning cognitive learning, psychomotor learning, and affective learning.

- What are some examples of learning you have experienced in each of the three categories?
- Certainly, you have acquired new knowledge (cognitive learning) and developed dexterity in some areas such as typing or sketching or dancing (psychomotor learning). Do you have more difficulty coming up with examples under the affective learning category?

ABET - Engineering Criteria:

Review the following 9 attributes presented that the ABET Engineering Criteria mandates for engineering graduates.

- Which primarily involves cognitive learning?
- Which is most likely to involve psychomotor learning?
- What psychomotor skills might you acquire?
- Which of the attributes are most likely to involve affective learning?

Self-Reflection on Learning Style:

- When you receive new knowledge, what type of information do you prefer?
- Would you describe yourself as a sensing learner or an intuitive learner?

- Do you prefer to receive information visually or verbally?
- Which of the common mistakes made by freshman students have you made yourself or believe you might make? What steps will you take to avoid making them in the future?

Reflection and transparency

Prepare a brief reflection that includes:

- The AI tool used and how (If you used one)
- What you learned while doing this lab

Grading Rubric

- **Objectives:** Identifies objectives and provides the necessary background or context (0-3 points)
- **Results/Analysis:** Presents results clearly. Analyzes findings correctly, with proper calculations and/or comparisons as needed. (0-6 points)
- **Discussion & Reflection:** Explains significance of results, discusses errors and connects outcomes to engineering/technology concepts. (0-5 points)
- **Professionalism & Format:** Report is well-organized, follows the provided template, uses clear language, proper grammar and formatting. Includes citation and references (if outside sources were used) (0-6 points)
- **Total:** 20 points

Deliverables

- Submit your completed activity responses as instructed by your instructor.